

ECCB Tutorial Curriculum

Neural posterior estimation for population genetics

Duration: ~3 hours 25 minutes (+ open-ended Discussions)

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Schedule Overview

Time	Session	Duration
09:00	Opening Presentation	15 min
09:15	Part 1: Population Genetics	40 min
09:55	<i>Break</i>	5 min
10:00	Part 2: Simulation-Based Inference	70 min
11:10	<i>Break</i>	15 min
11:25	Part 3: Snakemake Workflow	60 min
12:25	Discussions	—

Opening Presentation (15 min)

"SBI in Population Genetics: Motivation, Methods, and Results"

- Introduction to population genetics and why we care about inferring demographic parameters
 - Classical likelihood-based approaches and their limits
 - What is simulation-based inference (SBI)?
 - Neural Posterior Estimation (NPE) and competing algorithms (e.g., ABC)
 - The role of summary statistics as the bridge between data and inference
 - Showcase: how we implemented and evaluated SBI for inferring N_e and recombination rates
 - Data pipeline: msprime coalescent simulations + summary statistics (SFS, LD)
 - Training strategy and validation
 - Key results in our [preprint](#)
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Part 1: Population Genetics (40 min)

Showcase in Notebook: [1_data_simulation_msprime.ipynb](#)

Goal: build intuition about coalescent simulations and summary statistics before touching any inference machinery.

1.1 The coalescent model (10 min)

- Genealogies, effective population size N_e , mutation and recombination rates
- How `msprime` encodes these as tree sequences

1.2 Simulating a constant-size population (15 min)

- Running `msprime.sim_ancestry()` and `msprime.sim_mutations()`
- Inspecting tree sequences and genotype matrices

1.3 Summary statistics: the site-frequency spectrum (SFS) (10 min)

- Extracting the allele frequency spectrum with `tskit`
- Folded vs. unfolded SFS
- Visualising and interpreting AFS shape under different N_e values

1.4 Q&A / buffer (5 min)

Break (5 min)

Part 2: Simulation-Based Inference (70 min)

Showcase in Notebooks: [2_introduction_to_sbi.ipynb](#), [3_sbi_in_popgen.ipynb](#), [4_playground_complex_scenario.ipynb](#)

Goal: understand NPE from a simple toy example up to a realistic popgen application, and appreciate how prior design shapes results.

2a. Introduction to SBI (20 min) — [2_introduction_to_sbi.ipynb](#)

Why SBI? (5 min)

- Intractable likelihoods and the role of simulators
- NPE, NLE, NRE at a glance — why this workshop focuses on NPE

The `sbi` toolkit: linear Gaussian example (10 min)

- Defining a `Prior` and a `Simulator`
- Training a neural density estimator with `NPE`
- Drawing posterior samples and visualising with `pairplot`

Posterior predictive checks (PPC) (5 min)

- Comparing simulated data from posterior draws to the observed data
- Reading a PPC plot: what good and bad coverage look like

2b. SBI in Population Genetics (30 min) — [3_sbi_in_popgen.ipynb](#)

Applying NPE to popgen (10 min)

- The full pipeline: msprime simulator → SFS summary statistic → prior → NPE
- Jointly inferring N_e and the recombination rate

Training and evaluation (15 min)

- Generating (θ, x) training pairs
- Training the density estimator and monitoring convergence
- Posterior predictive checks on held-out test data

Simulation-based calibration (SBC) (5 min) — *optional*

- Why calibration matters for scientific credibility
- Running SBC and interpreting rank histograms

2c. Playground — Complex Demographic Scenarios (20 min) —

[4_playground_complex_scenario.ipynb](#)

Multi-epoch demography (8 min)

- Designing a **BoxUniform** prior over epoch-specific N_e values and recombination rate
- Building piecewise-constant demography in **msprime**

Comparing demographic scenarios (7 min)

- Medium, Large, Decline, Expansion, Bottleneck, Zigzag
- How AFS shape encodes demographic history — and where it becomes ambiguous

Why your prior matters (5 min)

- Coverage, identifiability, and pathological priors
- Open exploration: participants modify priors and observe effects

Break (15 min)

Part 3: Snakemake Workflow (60 min)

Demo / Walkthrough | Notebook: [5_snakemake_workflow.ipynb](#) + **popgen-npe** pipeline

Goal: see how a research-grade SBI workflow is structured for reproducibility and scaling, and walk through the popgen-npe [Soup-to-Nuts tutorial](#) to infer a recombination rate landscape from a VCF.

3.1 Motivation: from notebook to pipeline (5 min)

- Why interactive notebooks don't scale: memory, parallelism, reproducibility
- What Snakemake brings: rule-based DAGs, checkpointing, cluster integration

3.2 Pipeline architecture (10 min)

- Repository structure of `popgen-npe`: `workflow/training_workflow.smk`, `workflow/prediction_workflow.smk`, `workflow/config/`
- The simulator → processor → embedding network → normalising flow pipeline
- One YAML drives both training and prediction (see `popgen_npe_demo/config.yaml`)

3.3 Live demo: running the pipeline (30 min)

- Walk through the Soup-to-Nuts tutorial's six steps inside `5_snakemake_workflow.ipynb`
- Steps 1–3 (simulator, processor, YAML): read the config that produced the shipped checkpoint
- Step 4 (training): explain, **do not** re-run live — checkpoint is shipped at `notebooks/popgen_npe_demo/checkpoint/`
- Step 5 (prediction): live run of `snakemake --snakefile workflow/prediction_workflow.smk` against the bundled example VCF
- Step 6 (interpret): load per-window posteriors, plot the rate landscape

3.4 Scaling to real data (10 min)

- Customising the YAML's `cpu_resources` / `gpu_resources` and SLURM options for cluster runs
- Swapping simulators or processors without changing pipeline code
- Tutorial "Next steps" and the `popgen-npe` [contributor guide](#)

3.5 Q&A and wrap-up (5 min)

- Key resources and follow-up reading
- How to adapt the pipeline to your own simulator and summary statistics